CS & IT ENGINEERING

Operating System

CPU Scheduling

Lecture-05

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Recap of Previous Lecture

Topic Round Robin Algorithm

Topics to be Covered

Topic Multilevel Queue Scheduling

Topic Multilevel Feedback Queue Scheduling

Topic Questions on Scheduling

[MCQ]

#Q. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

GATE-2013

A This algorithm is equivalent to the first-come-first-serve algorithm

B This algorithm is equivalent to the round-robin algorithm.

C This algorithm is equivalent to the shortest-job-first algorithm

D This algorithm is equivalent to the shortest-remaining-time-first algorithm

$T = 3$

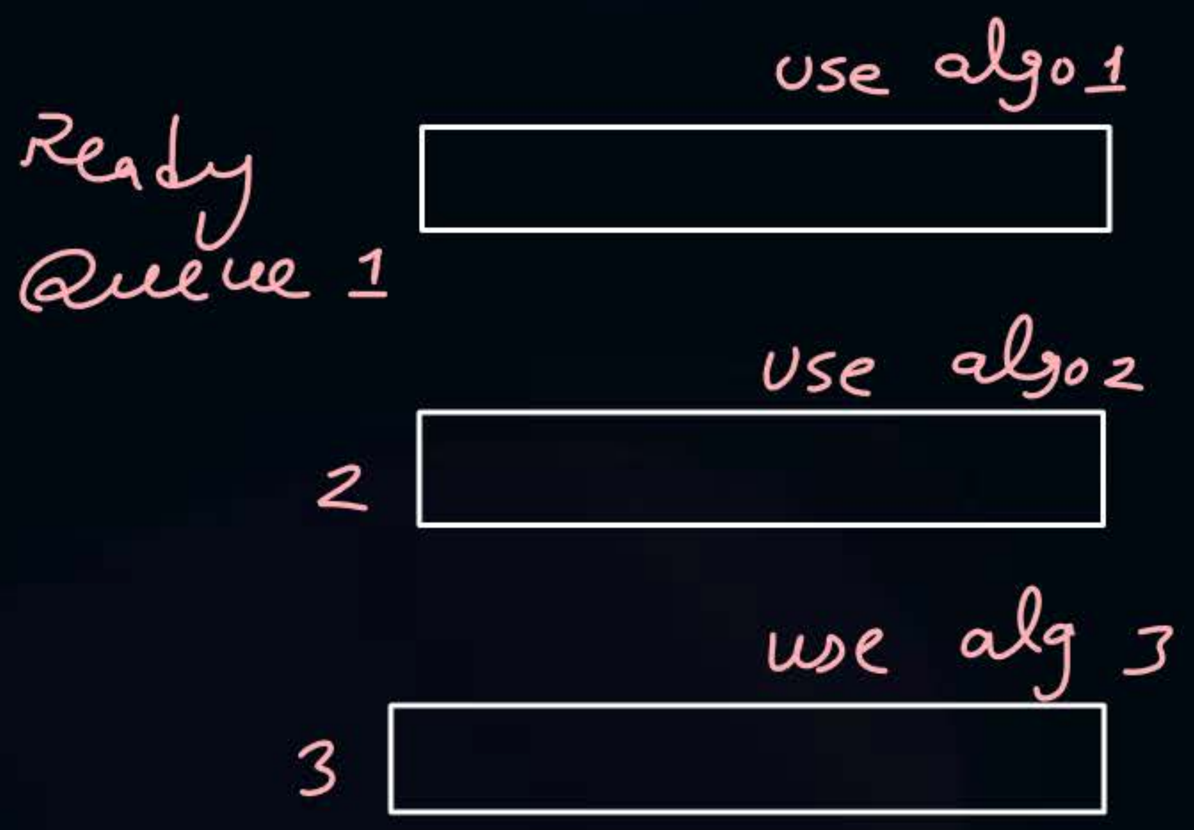
assume:-

	AT	Priority
P1	0	~~0~~ ~~3~~ ~~6~~ ~~9~~ 12
P2	0	~~0~~ ~~3~~ ~~6~~ ~~9~~ 12
P3	0	~~0~~ ~~3~~ ~~6~~ ~~9~~ 12

P1	P2	P3	P1	P2	P3	. . . .

0 3 6 9 12 15 18

Topic : Multilevel Queue Scheduling



Topic : Multilevel Queue Scheduling

Ready Queue

System Processes

Foreground Processes

Background Processes

Topic : Multilevel Queue Scheduling

1. Fixed priority preemptive scheduling method
2. Time slicing

Topic : Multilevel Queue Scheduling

Queue 1: RR with Q=2
(Higher Priority)

Queue 2: FCFS

Ready Queue 1 :- ~~P1, P2~~, P4

Queue 2 :- P3

Process	Arrival Time	Burst Time	Queue
P1	0	4	1
P2	0	3	1
P3	0	9	2
P4	9	4	1

P1	P2	P1	P2	P3	P4	P4	P3

0 2 4 6 7 9 11 13 20

Topic : Multilevel Queue Scheduling

Queue 1 :- ~~P1~~, ~~P2~~, ~~P4~~, ~~P6~~, ~~P4~~, ~~P7~~

Queue 2 :- P3, P5, P8

Queue 1: RR with Q=3
(Higher priority)

Queue 2: FCFS

P1	P2	P3	P4	P6	P4	P3	P7	P3	P5	P8
0–3	3–6	6–10	10–13	13–16	16–17	17–19	19–21	21–23	23–29	29–34

Process	Arrival Time	Burst Time	Queue
P1	0	3	1
P2	0	3	1
P3	2	8	2
P4	10	4	1
P5	11	6	2
P6	11	3	1
P7	19	2	1
P8	13	5	2

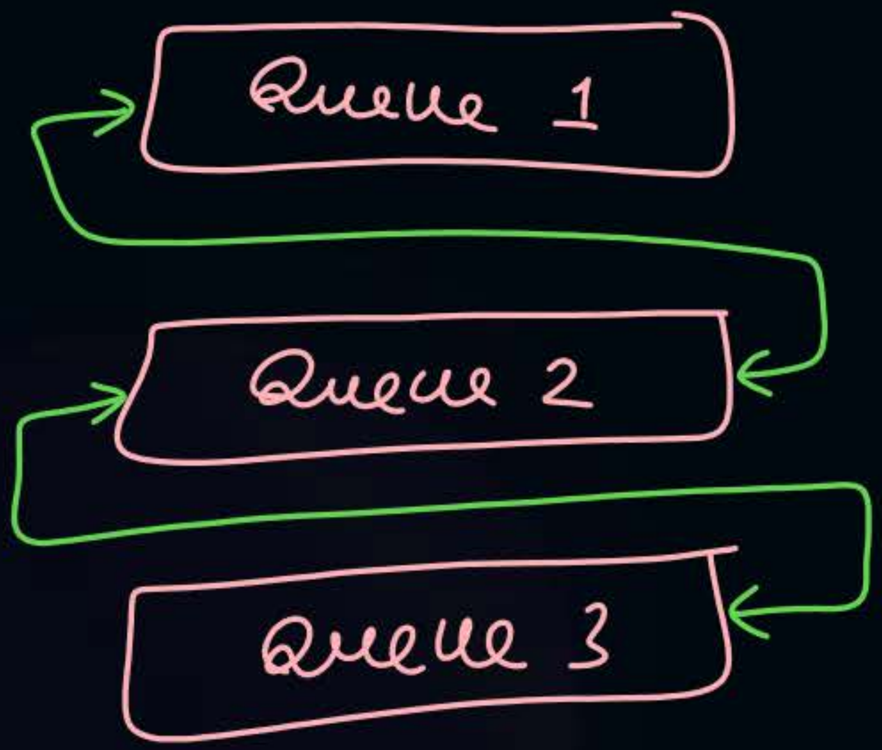
Topic : Multilevel Queue Scheduling

Disadvantages:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty

2. It is inflexible in nature. ⟶ Processes can not be shifted from one to another queue.

Topic : Multilevel Feedback Queue Scheduling



a process can be shifted from one Queue to another

Topic : Multilevel Feedback Queue Scheduling

Disadvantage:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty.

Advantage:

1. Flexible

[NAT]

H.W. ⇒ Practice

#Q. Consider the following set of processes:

Process	Arrival Time	Burst Time
P1	0	10ms
P2	0	29ms
P3	0	3ms
P4	0	7ms
P5	0	12ms

Calculate average waiting time for:
FCFS, Non-preemptive SJF, SRTF and Round-robin (quantum = 10ms)

Topic : Analysis of Scheduling Algorithms

Basis of Analysis	Algorithm
Minimum average WT among non-preemptive	SJF
Minimum average WT among all algos	SRTF
Non-preemptive always	FCFS, SJF, LJF, HRRN, Non-preemptive priority algo
Preemptive always	None
SRTF behaves as SJF	1. when all processes arrive together 2. when later arriving processes do not have BT lesser than Remaining time of running process.
Preemptive Priority behaves as Non-Preemptive	1. when all processes arrive together 2. when later arriving processes do not have priority higher.

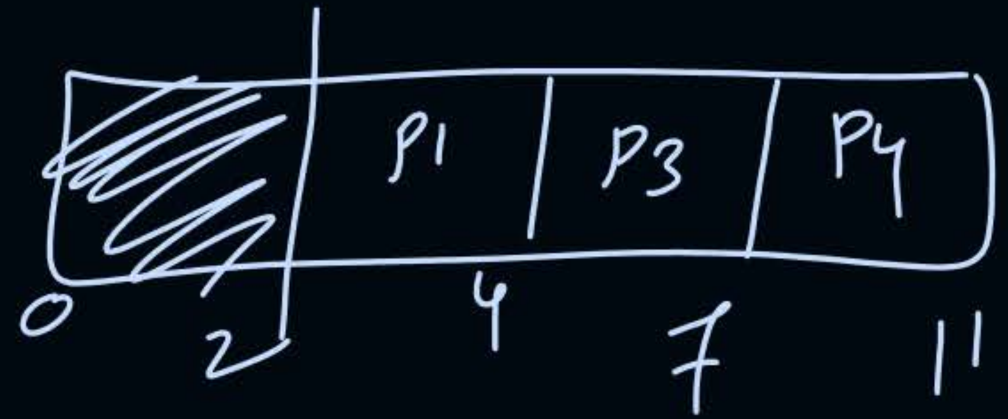
Topic : Analysis of Scheduling Algorithms

Basis of Analysis	Algorithm
RR behaves as Non-Preemptive	when $Q \geq$ Max (Burst time of all processes)
Convoy Effect	FCFS, LJF
Starvation	SJF, SRTF, Priority-based, LJF, LRTF

ex:-

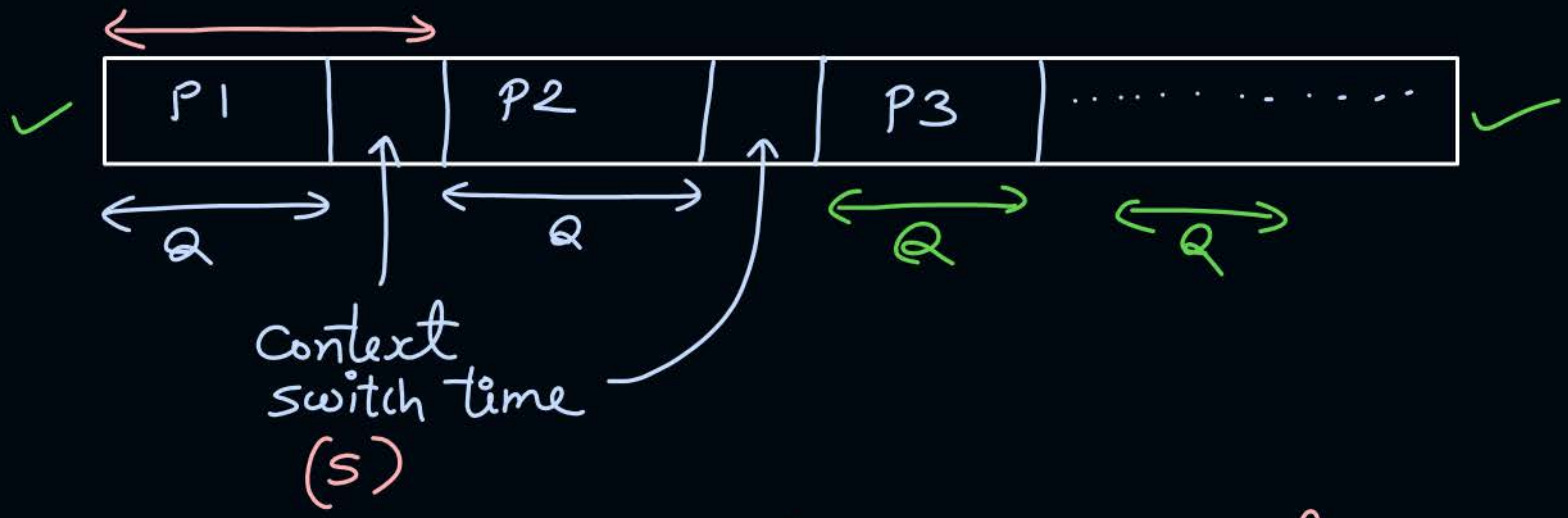
	AT	BT
P1	~~0~~ 2	2
P2	~~0~~ 2	4
P3	~~0~~ 2	3

SJF or SRTF



		BT
P1	0	10
P2	1	20
P3	2	30

In round robin:-   Quantum $= Q$



for each process execution:-

$$\text{efficiency of CPU} = \frac{Q}{Q+S} * 100\%$$

for n times process execut^n:-

$$= \frac{nQ}{nQ+(n-1)S} * 100\%$$

[NAT] **Homework**

#Q. Consider the following set of processes:

Process	Arrival Time	Burst Time
P1	0	10ms
P2	0	29ms
P3	0	3ms
P4	0	7ms
P5	0	12ms

FCFS = 28

SJF =13

SRTF =13

RR =23

Calculate average waiting time for:
FCFS, Non-preemptive SJF, SRTF and Round-robin (quantum = 10ms)

[MCQ]

#Q. Consider four processes P, Q, R, and S scheduled on a CPU as per round-robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time t = 0. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

GATE-2022

(A) P = 4, Q = 10, R = 6, S = 2

(B) P = 2, Q = 9, R = 5, S = 1

(C) P = 4, Q = 12, R = 5, S = 4

(D) P = 3, Q = 7, R = 7, S = 3

	AT	
P	0	≤4
Q	0	9 to 12
R	0	5 to 8
S	0	1 to 4

1 to 4

P	Q	R	S	Q	R	Q

←4→←4→ ←4→ 1 to 4

1 to 4

2 mins Summary

Topic Multilevel Queue Scheduling

Topic Multilevel Feedback Queue Scheduling

Topic Questions on Scheduling

Happy Learning

THANK - YOU